BTEC Level 3 Applied Science · Unit 1 Chemistry · Answer sheet

Covalent Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. A shared pair of electrons attracted to both nuclei.
2. An outer-shell pair not used in bonding.
3. e.g. H_2O or CO_2; diamond, graphite or SiO_2.
4. Each Cl shares one electron; the shared pair is attracted to both nuclei.
5. Each C bonded to four others in a tetrahedral giant covalent network.
6. Graphite has delocalised electrons between layers; diamond's electrons are all in C–C bonds.

7. 1, 2, 3 and 4 shared pairs.

8. Water is simple molecular (weak intermolecular forces). SiO_2 is giant covalent.

9. Boiling overcomes intermolecular forces; covalent bonds inside molecules remain.

10. Each H has 1 electron. They share a pair. Both then have a full first shell. [3]

11. Iodine is simple molecular — weak forces between I_2. Diamond is giant covalent. [4]

12. No mobile ions or delocalised electrons. [2]

13. Nitrogen donates its lone pair to H^+ (dative bond). All four N–H bonds in NH_4^+ are then equivalent.